

Cross-Cultural Surveys

Cognition, Communication and Culture

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Cross-Cultural Surveys – 1

- Single or multiple surveys involving multiple “cultural” groups
- Relevance with surveys in multinational, multiregional, and multicultural contexts (“3MC” surveys)
- Rapid growth through WWII
- Added layers of complexity to error and operations of survey research, compared to uni-cultural surveys

Cross-Cultural Surveys – 2

- Ideally designed to be comparative for the goal of comparing multiple groups across nations, regions or cultures
- Regardless of the design deliberation, survey data are widely used to compare populations (e.g., US vs. Canada) or population subgroups (e.g., US Whites vs. US Latinx)

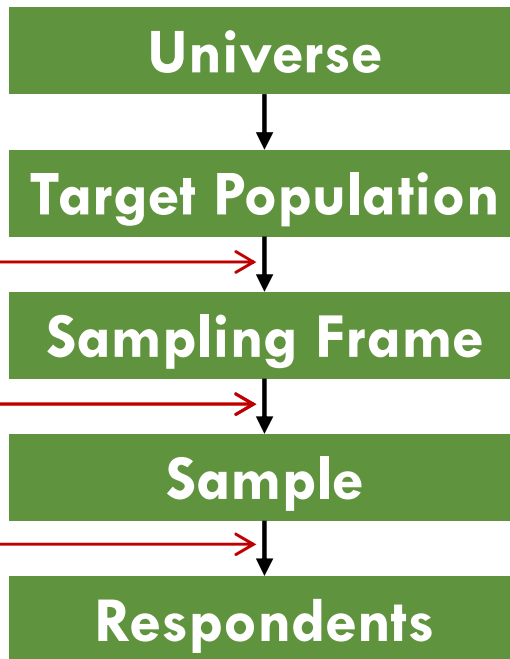
Cross-Cultural Surveys: Examples

	Social	Health	Educational
Global	International Social Survey Programme (ISSP) Comparative Study of Electoral Systems (CSES) World Values Survey (WVS)	World Mental Health Survey (WMH) World Health Surveys (WHS) Demographic and Health Surveys (DHS)	Programme for International Student Assessment (PISA)
Regional	European Social Survey (ESS) European Values Study (EVS) Arab/Asian/Afro Barometer	Survey of Health, Aging, and Retirement in Europe (SHARE)	
US	General Social Survey (GSS)	Health and Retirement Study (HRS) National Health and Aging Trends Study (NHATS)	National Assessment of Educational Progress (NAEP)

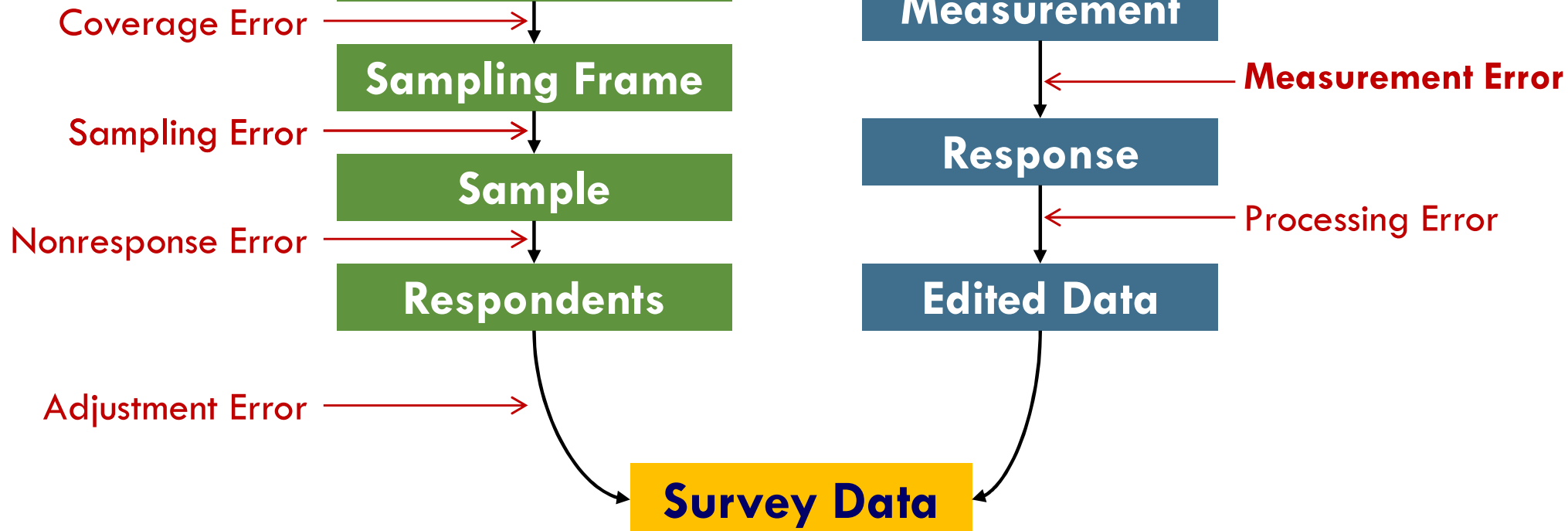
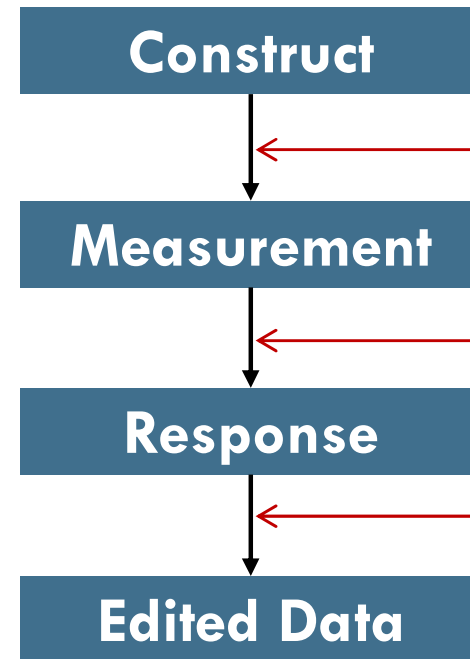
Total Survey Error

Groves et al. (2011)

Representation



Measurement



Adaptation and Translation

- Questions/questionnaire are developed in one particular survey context (e.g., one population, one mode, one language, one culture)
- Adaptation: deliberate modification of a question or questionnaire to reflect survey contexts
- Translation: From “source” to “target” language
- Example:
 - Translation: English to French
 - Adaptation: French-French vs. Canadian-French
 - “How proud are you to be [nationality]?”

Adaptation: Background – 2

- Not necessarily unique for 3MC
 - e.g., questions for Web vs. telephone mode
- Needs may emerge during translation or pretesting
 - Impossible to identify a clear-cut unique stage in the survey life cycle

Adaptation: Rationales – 1

- For measurement units
 - E.g., Metric vs. imperial
- To improve respondent comprehension
 - E.g., Fluzone Intradermal vaccine vs. flu shot
- To improve conceptual coverage
 - E.g., Vegetables: green salad, bok choy, komatuna

Adaptation: Rationales – 2

- To respect cultural discourse norms
 - E.g., Korean with a honorific system
- To accommodate cultural sensibilities
 - Culturally anchored norms (e.g., writing directions)
- To reflect lexicon and grammar
 - E.g., Neither agree or disagree
- To maintain/reduce difficulty

Adaptation Example

- Agreement scale response categories frequently use “neither agree nor disagree”
- In languages such as Hebrew :
 - Closest semantic option available to translate ‘disagree’ in Hebrew, for example, corresponds to ‘no agree’
 - ‘neither’ and ‘nor’ are the same as the target language element corresponding to ‘no’
 - If translated into “no agree, no no agree: makes little sense in Hebrew
 - ‘neither agree nor disagree’ translated into ‘in the middle’ (Harkness, 2003)

Adaptation Steps

1. Determine the policy, people, procedures for adaptation for the project.
2. Recruit a team for adaptation.
3. Review the source questionnaire for adaptation needs.
4. Review the translated questionnaire.
5. Document adaptations and the rationale.
6. Test adaptations with the target population.

Translation: Background – 1

- From source language to target language
- Unique to and central for multilingual surveys
- English as a source language
 - Largest lexicon of any language
- Part of the target language instrument development
- Good translation → comparability
- Bad translation → failure in comparability

Translation: Background – 2

- Translation in multilingual surveys requires
 - Understanding of translation,
 - Understanding of survey practice,
 - Understanding of survey measurement, and
 - Writing good questions
- Concept in the question may not be salient or relevant in all cultures
 - E.g., Political efficacy
 - Translation cannot solve this problem

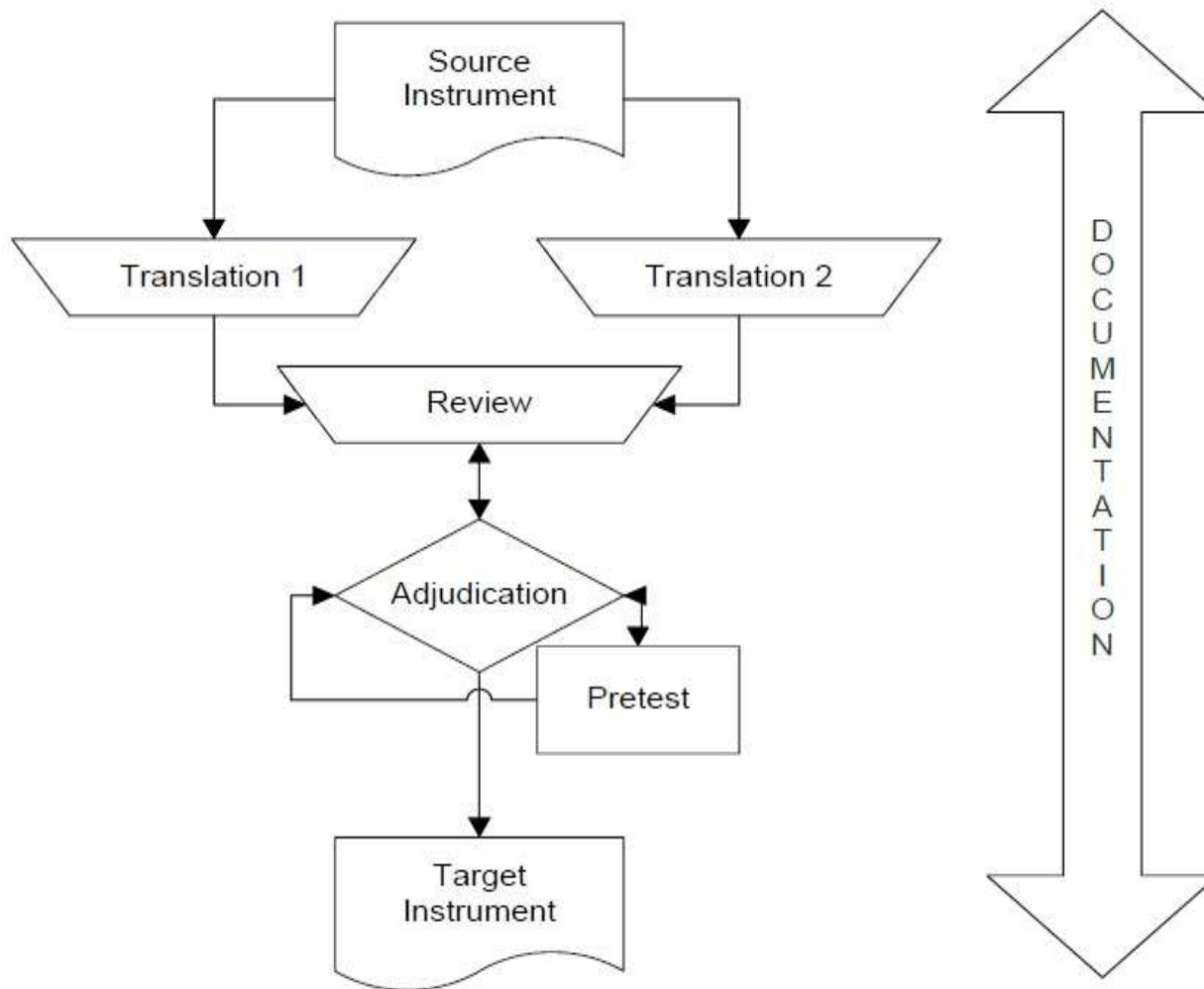
Translation: Background – 3

- Machine / AI translation
- Ad hoc do-it-yourself translation
- Unwritten “on the fly” translation
- Back translation
 - Process of re-translating content from the target language back to its source language in literal terms
- Team translation

Translation: Team Translation

- Translators – Reviewers - Adjudicator
- Partially iterative procedure
- Decision on when pretest and when finalizing translation for fielding
- Translation, Review, Adjudication, Pretesting and Documentation (TRAPD) Model

TRAPD Translation Model



- Reduces fixation on a single translation output
- Ensures a review process
- Avoid idiosyncrasies
- Avoids uneven lexicon

Cross-Cultural Surveys – 4

- Where does it stand?
 - Operational developments: Comparative by design
 - E.g., International Social Survey Programme, European Social Survey, Health and Retirement Study and its sister surveys
 - Methodological developments
 - “... progress exists in a specific – although somewhat frustrating – way” (Scheuch, 1968)
 - Survey methodologist, Survey researchers and Substantive researchers

Cross-Cultural Surveys – 5

- Why?
 - Difficult
 - Advance in survey methodology is
 - Driven mostly by Western scholars or Western-trained scholars
 - Based on studies mostly on Western populations
 - Definition of culture in survey methodology
 - Nationality, race, ethnicity as “culture”
 - Often a control variable or a correlate

Cross-Cultural Surveys – 6

- How to deal with difficulties?
 - Better design
 - Data collection harmonization
 - Pretesting, Cognitive interviews, Experiments
 - Limitations
 - Costs and time
 - Not always feasible to deliberately design for comparability

Cross-Cultural Surveys – 7

- How to deal with difficulties? (cont'd)
 - Adjustment models
 - Borrowed from measurement theories
 - Limitations
 - Start with attitudinal unmeasurable constructs
 - Require additional data collection
 - Modeling assumptions
 - Measurement theories may not reflect realities of survey interviews

What is “Culture”?

- Many definitions
- Experience/Knowledge learned and shared among a group of people (Goodenough, 1971; Hong, 2009; Shweder & LeVine, 1984)
- Software of the mind (Hofstede)
- Culture is to society what memory is to individuals (Triandis, 1994)
- Dynamic and situational
- Nationality, race, ethnicity and language – powerful agents of culture (Hong, 2009)

Characterizing Culture – 1

- Cultural dimensions (Hofstede)
 - A framework for cross-cultural communication
 - Describes effects of a society's culture on the values of its members, and how these values relate to behavior
- Cultural syndromes (Triandis)
 - Consist of shared attitudes, beliefs, norms, role and self definitions, and values of members of each culture that are organized around a theme
 - e.g., individualism/collectivism
 - e.g., emotional expression/suppression
- Cultural norms
 - Rules or expectations of behavior and thoughts based on shared beliefs within a specific cultural or social group
- Cultural values
 - Preferences people have for how they navigate life
 - Indicate how people prefer to interact, communicate, plan, and complete tasks.

Characterizing Culture – 2

6 orthogonal cultural dimensions by Hofstede

1. Individualism vs. Collectivism

Degree to which people in a society are integrated into groups

2. Power distance: Small vs. Large

Extent to which the less powerful members of a society accept/expect that power is evenly distributed

3. Uncertainty avoidance: Weak vs. Strong

Society's tolerance to ambiguity

4. Masculinity vs. Femininity

Men should & women may be assertive/ambitious vs. both sexes modest/caring

5. Long-term vs. Short-term orientation

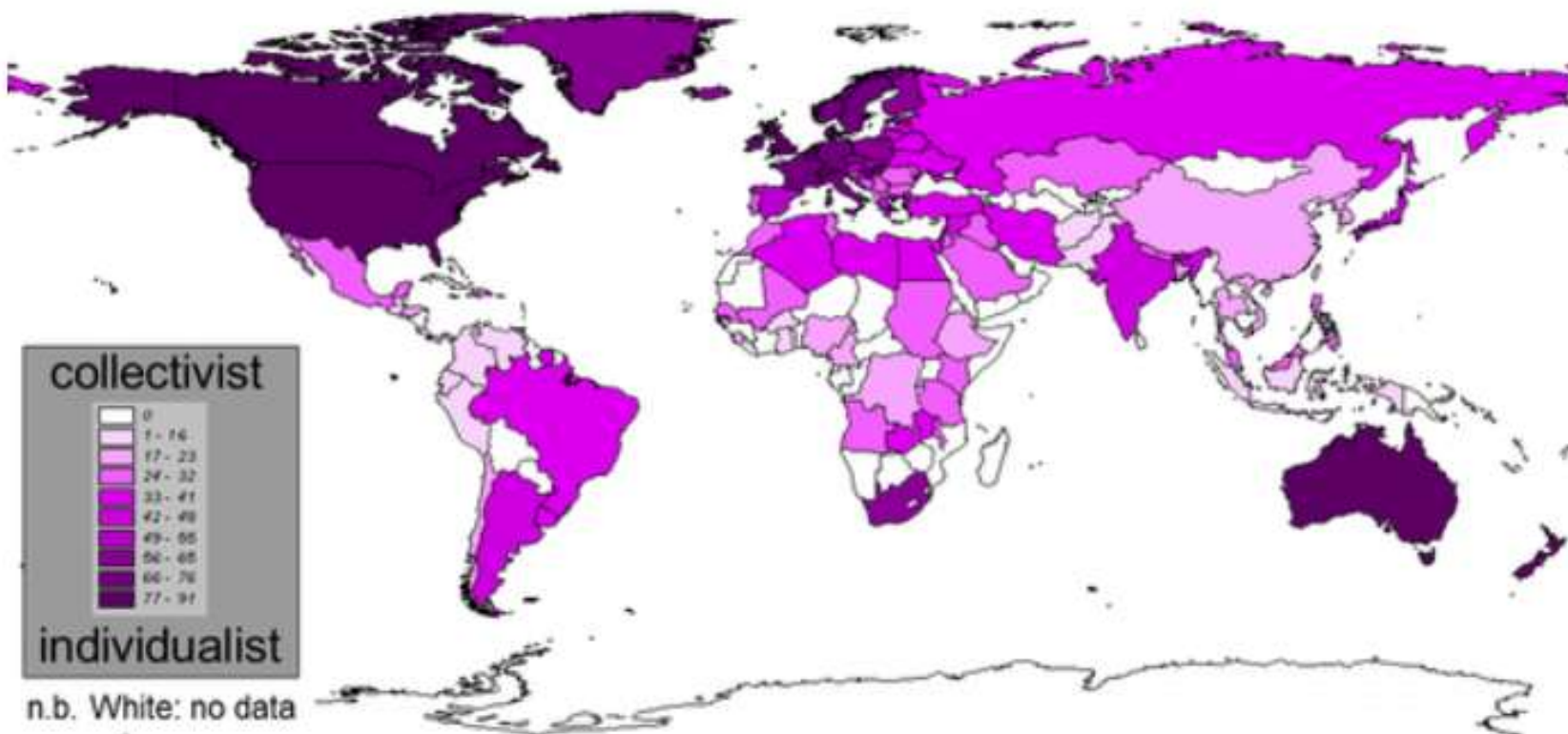
Important life events will occur in the future vs. in the past or now

6. Indulgence vs. Restraint

High vs. low importance on leisure

Characterizing Culture – 3

Collectivism – Individualism World map



Hofstede's model characterizes countries (<https://geerthofstede.com/culture-geert-hofstede-gert-jan-hofstede/6d-model-of-national-culture/>)

Characterizing Culture – 4

- Other culture-variant factors influencing cognition
 - Time orientation: Past, present and future
 - Use of context: High vs. low context
 - Space needs
 -
- “Silent language” (Hall)
 - "The Silent Language shows how cultural factors influence the individual behind his back, without his knowledge." —Erich Fromm

Characterizing Culture – 5

- Why examine culture this way?
 - Culture as a dynamic phenomenon and as shared knowledge
- Why relevant for survey research?
 - Describing vs. explaining culture

Cognition, Communication, and Culture – 1

- Culture as learned/shared knowledge and experience
- Verbal/nonverbal communications required
- Language used in conversations and survey interviews alike is shown to bring its cultural norms to the discourse (Lee & Schwarz, 2014; Ross, Xun, & Wilson, 2002)
 - Grammar and vocabulary influence the level of importance of elements of a given event in cognition

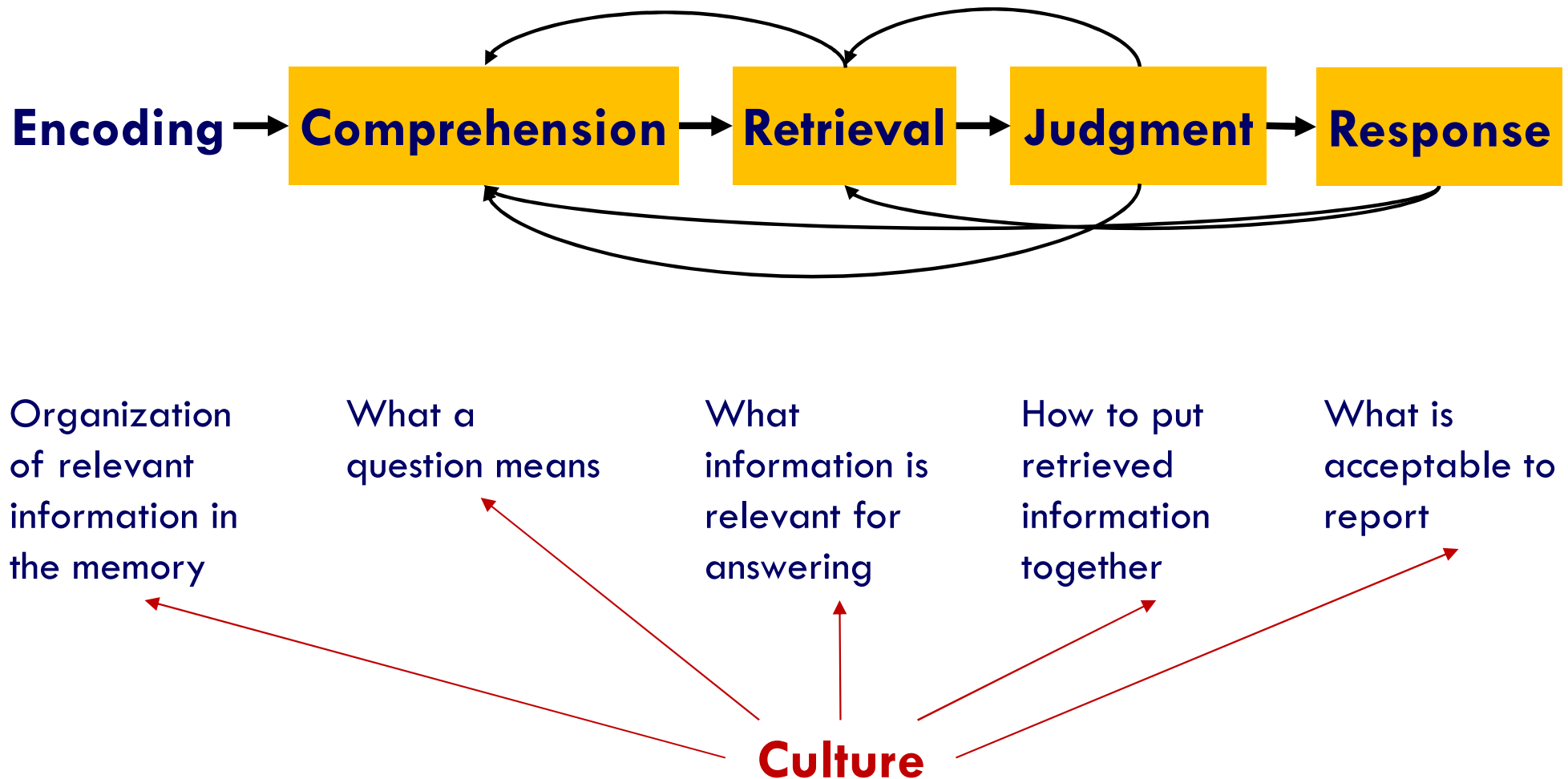
Cognition, Communication, and Culture – 2

- Culture provides a systematic framework for individuals to organize personal experiences → Culture is one of the elementary constructs of cognition (Nisbett, Peng, Choi, & Norenzayan, 2001)
E.g.,
 - Individualism vs. collectivism
 - Time orientation
- Culture matters to survey measurement through the role of cognition in survey response process

Between Group Differences

Dimension	Collectivism	Individualism	Honor
Focus...	on the group, with individuals conceived as parts of the group, connected to each other	on the individual as distinct and separate from others, with groups seen as made up of individuals	On the trustworthiness of individuals and groups with each depending on the other to prevent loss of reputation
Fosters...	Preference to fitting in	Preference for uniqueness	Sensitivity to reputation
Mental Procedure..	Connecting, emphasizing the embedeness of a stimulus in its field	Pulling stimulus apart from the field, segmenting	Ranking
Cultural Universal	because people do not survive alone, some mechanism of recognizing and feeling obligated to one's group is needed in any society	Societies need innovation to grow and flourish, which requires permitting members to at least sometimes innovate, do something different	

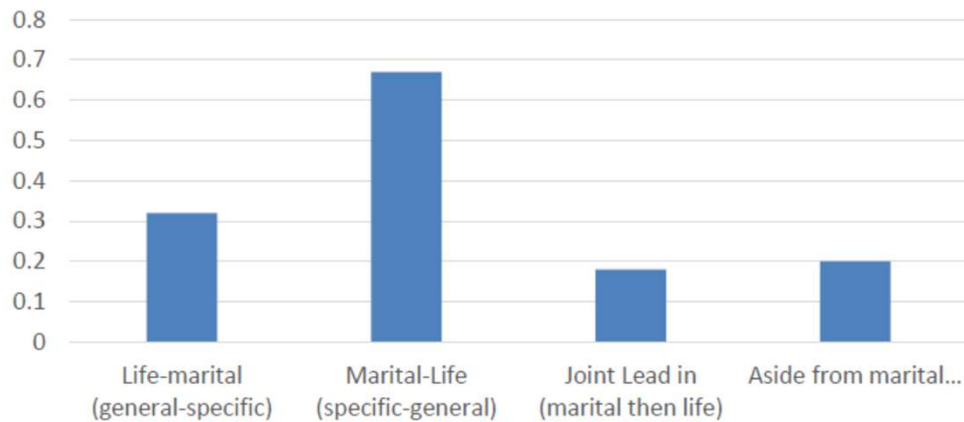
Response Process and Culture



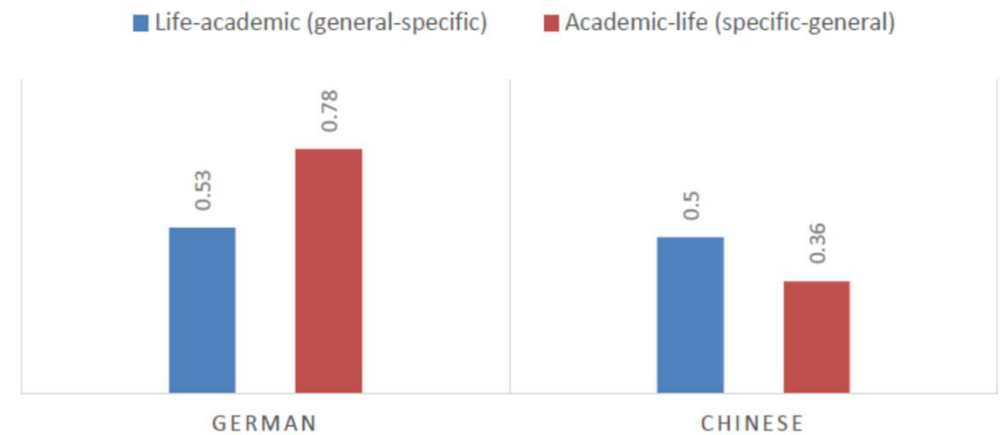
Culture matters to survey measurement through the role of cognition in survey response process

Comprehension

Correlation Between Responses to Two satisfaction Questions Among German Community Sample (Schwarz et. al, 1991)



Correlation Between Responses to Two satisfaction Questions Among Student Samples (Haberstroh, 2002)



- Collectivistic respondents are
 - more attentive to the context → Increases pragmatic differences in interpreting the second question
 - more likely to consider background information about the questioner that may bear on the likely common ground
- Translation cannot solve this problem
- Respondents can be primed

Comprehension

- Pragmatic meaning of the question
- Nonredundancy
 - Attention to the common ground
 - Don't repeat what the listener already knows
 - Redundant questions are interpreted as asking for new information
 - General → Specific question sequence

Retrieval – 1

- Culture may influence what people attend to and how they organize information leading to differences at encoding stage and/or influence what people attempt to retrieve and how they organize the retrieved information at recall or reporting stage
- Combination of how information is stored and how it is cued for recall

Retrieval – 2

- Collectivist cultures put a premium on “fitting” in
 - Need to monitor their own behavior as well as the behavior of others to avoid standing out
 - Apply to public events (more collectivistic vs. individualistic memories)
- Han et al. (1998): American vs. Chinese children recalled different events
 - American made more reference to self; Chinese made more reference to others
 - American made reference to more internal states and emotions; Chinese made reference to context of event

Judgment

- How to put retrieved information together
- Collectivistic cultures attend to the context
- Lee & Schwarz (2014):
 - For English speakers, the question context effect was smaller as the positive health reports were in the range of 75.4% to 80.5%

	English Speakers		Spanish Speakers	
	n	% (SE)	n	% (SE)
Self-rated health asked before health questions (HRS)	16343	75.4 (0.5)	801	41.3 (2.2)
Self-rated health asked after health questions (NHIS)	12039	80.5 (0.4)	688	68.3 (2.2)

Response – 1

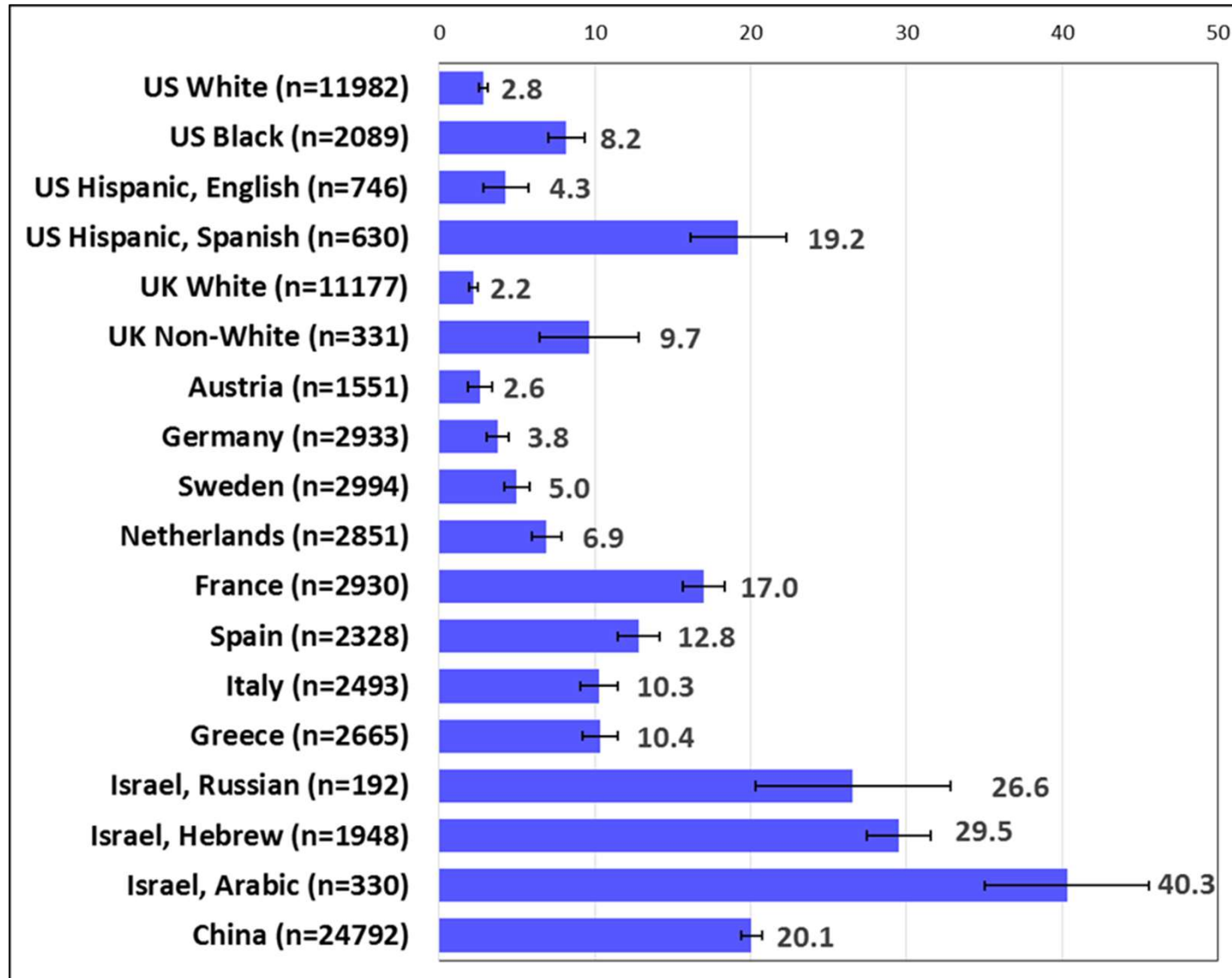
- Content considered favorable differs by cultures
- Individualist culture
 - Encourage a view of the self in a unique and positive terms; display more self-enhancement bias in the form of unrealistic unconscious positive views (Baumeister, 1998)
 - Value honesty in interaction with others (Triandis, 1995)
- Collectivist cultures
 - Emphasize maintenance of harmonious relationship; more concerned about fitting in, saving face, keeping honor
 - Limited “editing” of the truth is considered acceptable in the interest of maintaining honor, harmony, and saving face (Triandis, 1995) → More impression management

Response – 2

- Lee, et al. (2017)
- Subjective Life Expectancy: “What is the percent chance that you will live to be XX years old or more?”
- Difficult question
- Time orientation and its influence on cognition
 - Differential question difficulty by cultural groups
 - Differential item nonresponse due to “don’t know”

Response – 3

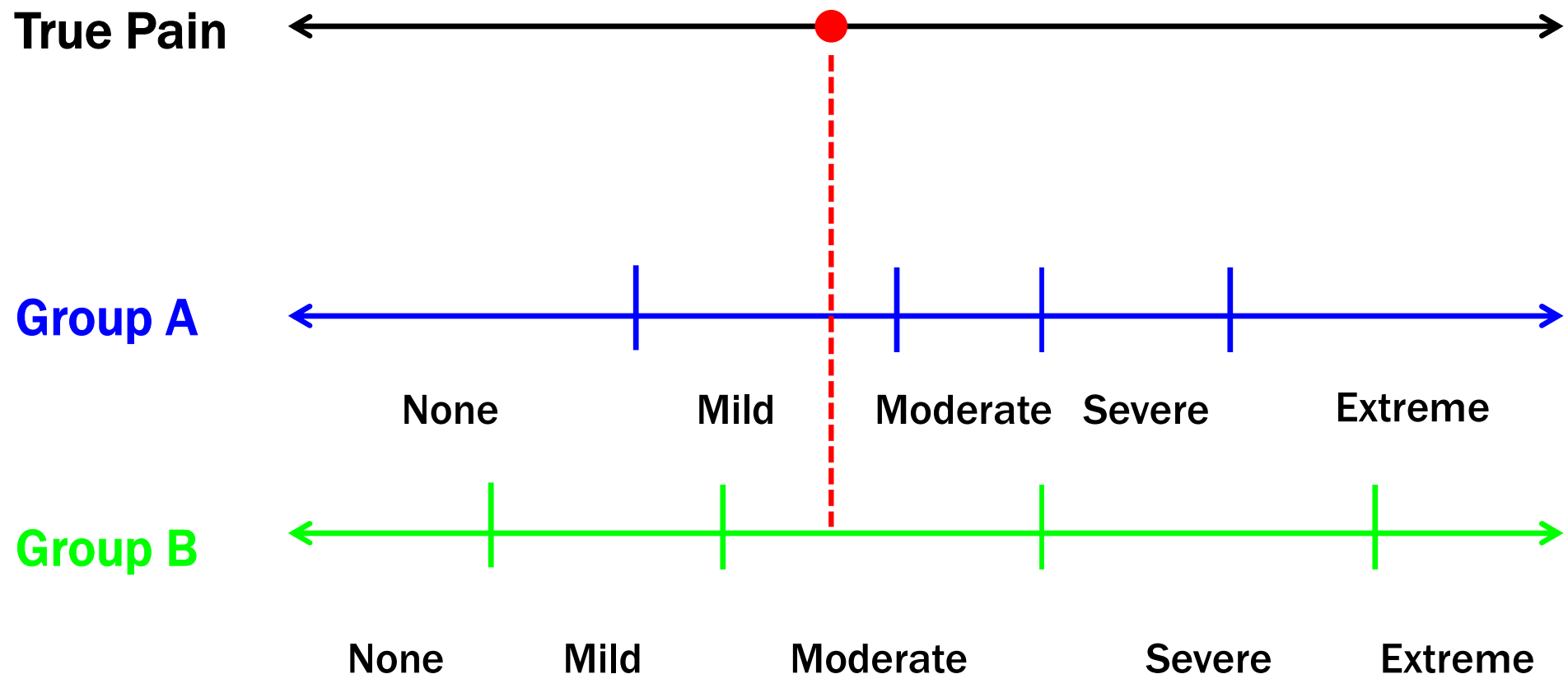
Item nonresponse rates on Subjective Life Expectancy



Measurement Error: Reporting

Heterogeneity

Q: Overall, in the last 30 days, how much pain or bodily aches did you have?



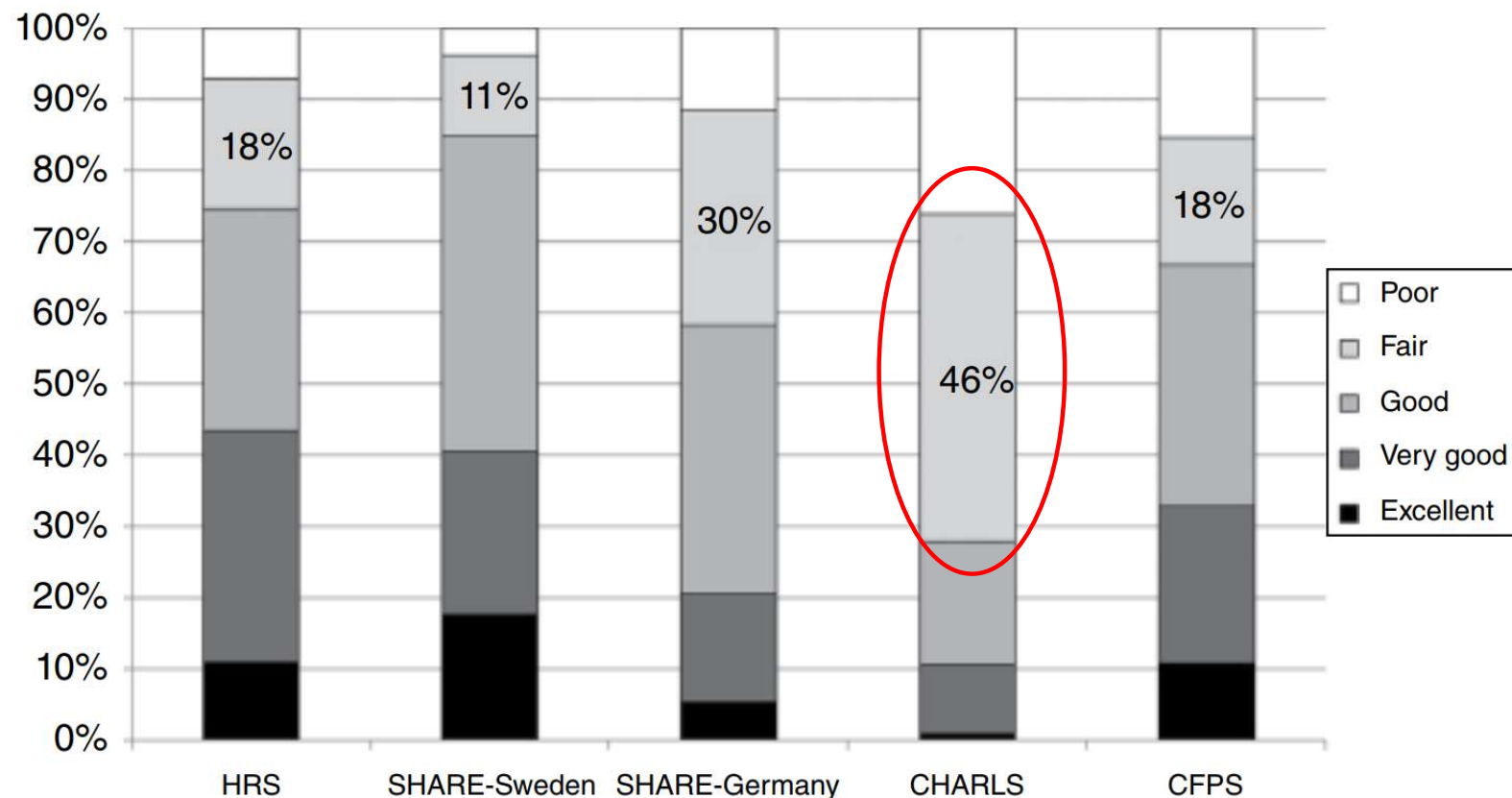
Adopted from Hu, Lee and Xu (2018).

Causes for Reporting Heterogeneity

- Individuals apply different standards in the selection of a response category
 - Respondents understand and use ordinal response categories in different ways.
 - Translation-triggered reporting heterogeneity.

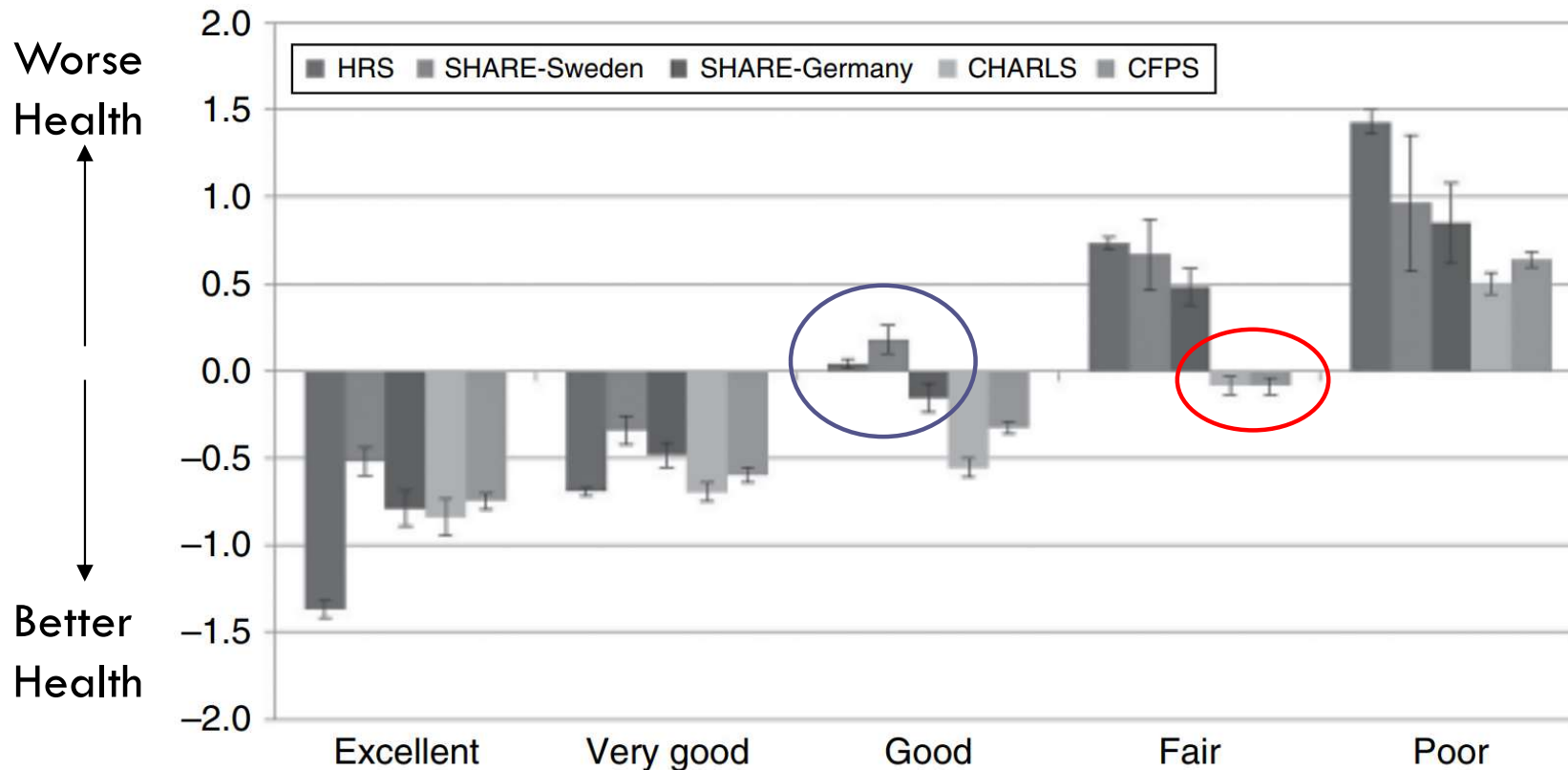
Reporting Heterogeneity Example

- Weighted response distributions of answers to self-rated health (Excellent to Poor scale)



Reporting Heterogeneity Example

- Weighted means of countries' standardized number of health issues by response categories of self-rated health



Yan, T., & Hu, M. (2018). Examining Translation and Respondents' Use of Response Scales in 3MC Surveys. *Advances in Comparative Survey Methods: Multinational, Multiregional, and Multicultural Contexts (3MC)*, 501-518.

Goals in Survey Measurements – 1

- Goal: Error-free measurement
- Simple measurement error model
$$y_i = \mu_i + \epsilon_i, \text{ where}$$
 - y_i : measured value of subject i ;
 - μ_i : “true” value of subject i ; and
 - ϵ_i : measurement error - Distribution of ϵ_i determines the nature of measurement error
- Function of y_i in the analysis
- E.g., Self-reported weight (y_1) and height (y_2)
vs. BMI ($z = f(y_1, y_2)$)

Goals in Survey Measurements – 2

- $y_{it} = \mu_i + \epsilon_{it}$
 - t : Time element;
 - y_{it} : Measured value of subject i at time t ;
 - μ_i : “true” value of subject i ; and
 - ϵ_{it} : measurement error at time t for subject i
- Measurement evaluation criteria for y :
 - $\text{Validity}(y) = \text{Cor}(y_i, \mu_i)$
 - $\text{Reliability}(y) = \frac{\text{Var}(\mu_i)}{\text{Var}(y_i)}$
 - $\text{Bias}(y) = E_i[E_t(y_{i,t}) - \mu_i]$

Survey Measurements for 3MC – 1

- Goal: Comparable measurement/error
- When groups A and B are involved,
 $y_{Ai} = \mu_{Ai} + \epsilon_{Ai}$ and $y_{Bi} = \mu_{Bi} + \epsilon_{Bi}$
- Comparability on the errors, ϵ_{Ai} and ϵ_{Bi} , determines measurement comparability
- Functional equivalence: the scale measures the same construct in every group
- Measurement invariance test can evaluate the equivalence of a measured construct in different groups across cultures

Survey Measurements for 3MC – 2

- $y_{Ai} = \mu_{Ai} + \epsilon_{Ai}$ and $y_{Bi} = \mu_{Bi} + \epsilon_{Bi}$
- Consider evaluation criteria (validity, reliability, etc.) for y_A and y_B separately
- Compare the criteria across groups A and B
- Functional equivalence (Scheuch, 1968)
 - Focusing on comparison error
 - Focusing on comparability
 - Comparability of scales

Survey Measurements for 3MC – 2

- What do we do?
 - Develop conceptual frameworks
 - Based on theories (e.g., cross-cultural psychology)
 - Incorporate culture in understanding survey response process
 - Focus on **equivalence**
 - Reflect the reality of survey data collection
- Until fully addressed, risk of analyzing y_A and y_B with blindly assuming $\epsilon_A = \epsilon_B$

Measurement Invariance

- Measurement invariance (MI) test: examine and ensure the comparability of multi-item scales across groups
- Multi-item scale: a set of differently worded question items related conceptually, developed to measure the same construct, and can be aggregated into a total score
- MI examines
 - Whether a scale measures the same underlying construct with the same structure across groups
 - Whether survey question items have the same meaning for male and female, elderly and teenagers, and people with different race/ethnicity and cultural backgrounds...

Multi-item Scale Example

- Center for Epidemiologic Studies Depression Scale (CES-D)
- 20 Items
- Measuring depression:
 - Scoring methodology for creating a summary score
 - Latent variable modeling, e.g., factor analysis

	During the Past Week			
	Rarely or none of the time (less than 1 day)	Some or a little of the time (1-2 days)	Occasionally or a moderate amount of time (3-4 days)	Most or all of the time (5-7 days)
1. I was bothered by things that usually don't bother me.	☐	☐	☐	☐
2. I did not feel like eating; my appetite was poor.	☐	☐	☐	☐
3. I felt that I could not shake off the blues even with help from my family or friends.	☐	☐	☐	☐
4. I felt I was just as good as other people.	☐	☐	☐	☐
5. I had trouble keeping my mind on what I was doing.	☐	☐	☐	☐
6. I felt depressed.	☐	☐	☐	☐
7. I felt that everything I did was an effort.	☐	☐	☐	☐
8. I felt hopeful about the future.	☐	☐	☐	☐
9. I thought my life had been a failure.	☐	☐	☐	☐
10. I felt fearful.	☐	☐	☐	☐
11. My sleep was restless.	☐	☐	☐	☐
12. I was happy.	☐	☐	☐	☐
13. I talked less than usual.	☐	☐	☐	☐
14. I felt lonely.	☐	☐	☐	☐
15. People were unfriendly.	☐	☐	☐	☐
16. I enjoyed life.	☐	☐	☐	☐
17. I had crying spells.	☐	☐	☐	☐
18. I felt sad.	☐	☐	☐	☐
19. I felt that people dislike me.	☐	☐	☐	☐
20. I could not get "going."	☐	☐	☐	☐

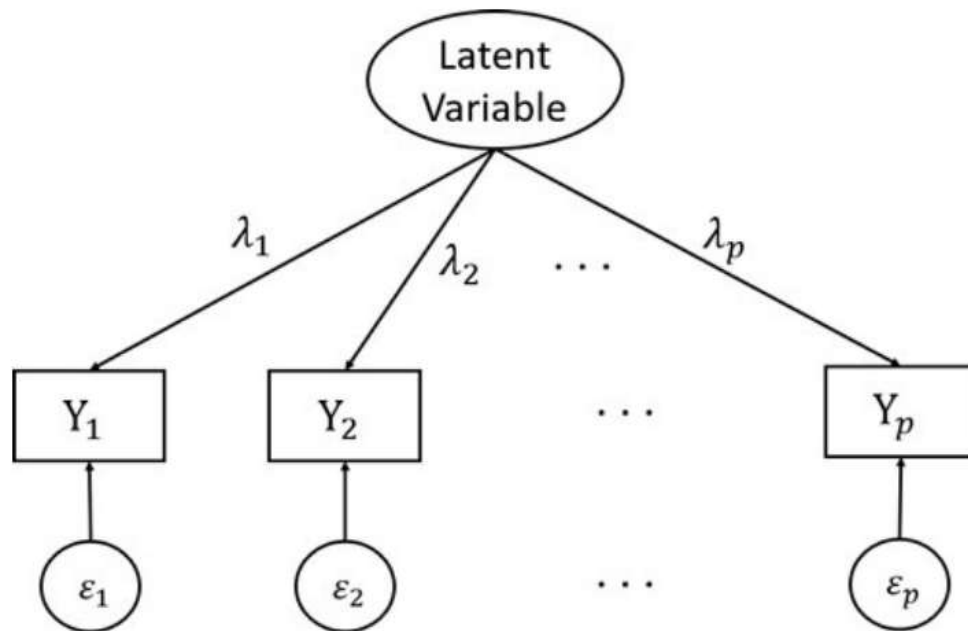
SCORING: zero for answers in the first column, 1 for answers in the second column, 2 for answers in the third column, 3 for answers in the fourth column. The scoring of positive items is reversed. Possible range of scores is zero to 60, with the higher scores indicating the presence of more symptomatology.

Multiple-Group Confirmatory Factor Analysis (MGCFA)

- Continuous, ordinal or binary distributed indicators
- Four levels of MI:
 - Configural invariance (least restrictive)
 - Metric / weak invariance
 - Scalar / strong invariance
 - Strict invariance (most restrictive)
- The first three levels are essential for latent mean comparisons across groups

MGCFA - Configural invariance

Figure 1. An illustration of a single-factor CFA with p indicators.

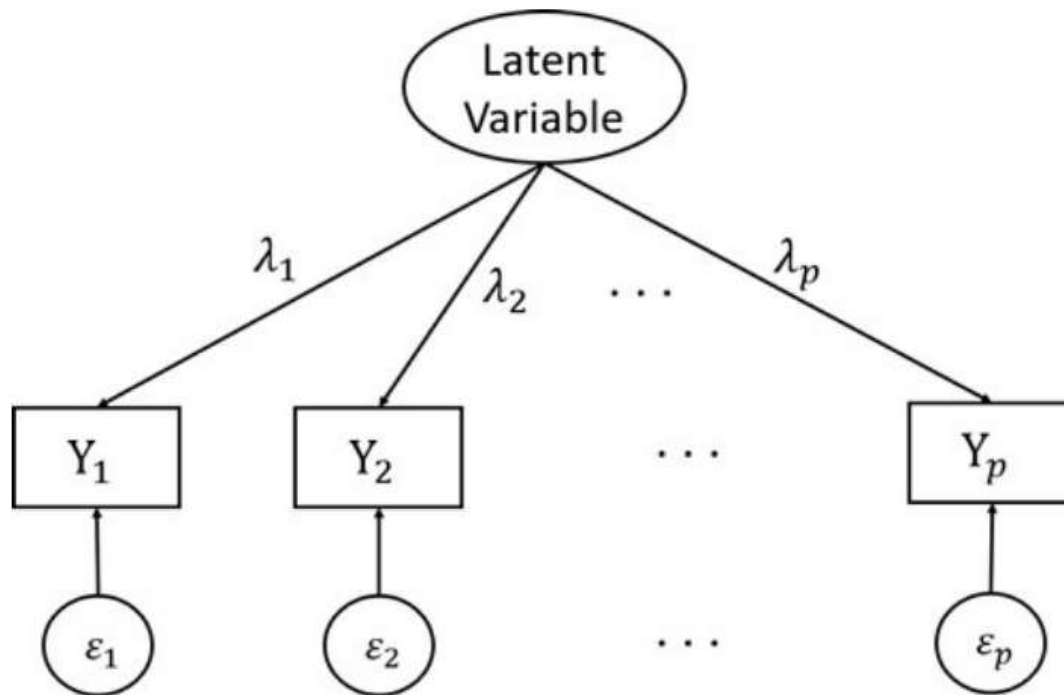


$$Y_g = \tau_g + \Lambda_g \eta + \varepsilon_g$$

- Y is a $p \times 1$ vector of scores on the p observed indicators
- τ is a $p \times 1$ vector of intercepts where p is the number of indicators
- Λ is a vector of factor loadings
- η is the latent score
- ε is a $p \times 1$ vector of residuals or error terms representing the unique portion of the observed variables not explained by the common latent variables
- g , group subscript, indicating that each of the parameters can differ by group

MGCFA - Metric invariance

Figure 1. An illustration of a single-factor CFA with p indicators.

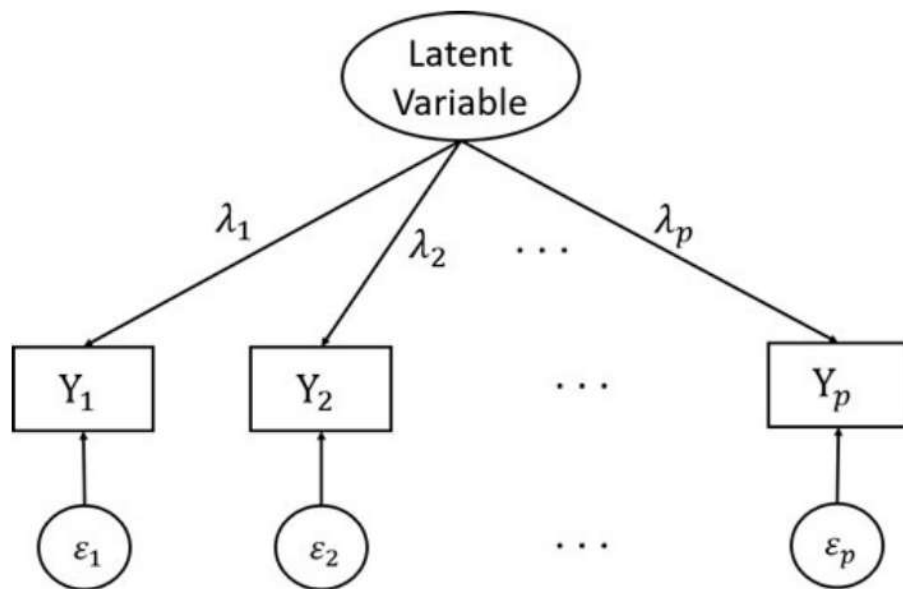


$$Y_g = \tau_g + \Lambda\eta + \varepsilon_g$$

- Requires the factor loadings (Λ) to be equal across comparison groups
- Full or partial metric invariance

MGCFA - Scalar invariance

Figure 1. An illustration of a single-factor CFA with p indicators.

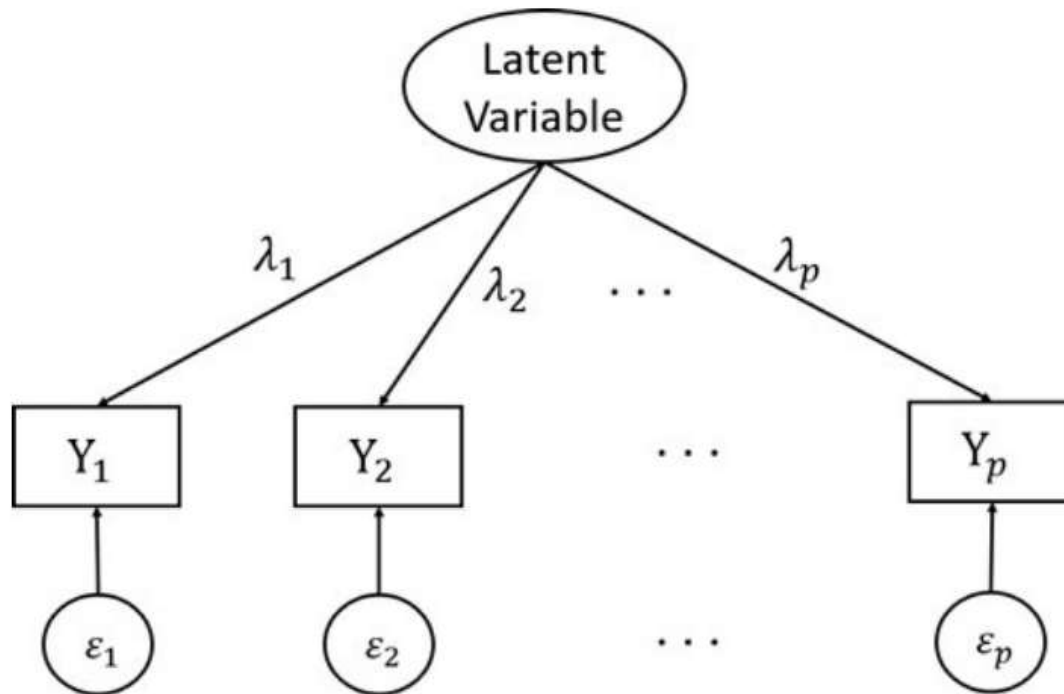


$$Y_g = \tau + \Lambda\eta + \varepsilon_g$$

- Requires not only the factor loadings but also the item intercepts (τ) to be invariant across groups
- Final level of MI-testing that is required to be able to compare latent means across groups
- Only with scalar invariance (at least partial scalar invariance), one can more confidently conduct latent means comparisons across groups

MGCFA - Strict invariance

Figure 1. An illustration of a single-factor CFA with p indicators.



$$Y_g = \tau + \Lambda\eta + \varepsilon$$

- Assess whether there is group invariance with respect to the unique residual variance (ε)
- Necessary in order to attribute group differences in the covariance structure (variances and covariances/correlations) of the observed indicators to the corresponding differences at the latent variable level

Summary

- Observed cultural differences in survey responses may be
 - True differences in attitudes or behaviors
 - Differences in response process
 - Differences in different methods used (such as mode)
 - Mixture of all
 - Long been recognized as an issue, but methodological research is relatively recent and limited
- Investigate potential reporting heterogeneity and evaluate measurement invariance before comparing across cultural groups!

Thank you!

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